Experiment No. 25 – Watch Recognition Using OpenCV

1. Aim

To recognize a watch from the given image using object recognition techniques in OpenCV.

2. Program

This program uses ORB feature detection and feature matching to identify a watch by comparing it with a reference watch image.

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload reference watch and test image
uploaded = files.upload()
names = list(uploaded.keys())

if len(names) < 2:
    print("Please upload two images: reference watch and test image.")
else:
    # Read images
    ref = cv2.imdecode(
        np.frombuffer(uploaded[names[0]], np.uint8),
        cv2.IMREAD_COLOR
    )

    test = cv2.imdecode(
```

```
    np.frombuffer(uploaded[names[1]], np.uint8),
    cv2.IMREAD_COLOR
)

if ref is None or test is None:
    print("Unable to load images")
else:
    # Convert to grayscale
    gray1 = cv2.cvtColor(ref, cv2.COLOR_BGR2GRAY)
    gray2 = cv2.cvtColor(test, cv2.COLOR_BGR2GRAY)

    # Create ORB detector
    orb = cv2.ORB_create(nfeatures=1000)

    # Detect keypoints and descriptors
    kp1, des1 = orb.detectAndCompute(gray1, None)
    kp2, des2 = orb.detectAndCompute(gray2, None)

    if des1 is None or des2 is None:
        print("No matching features found")
    else:
        # Match features
        bf = cv2.BFMatcher(
            cv2.NORM_HAMMING, crossCheck=True
        )
```

```
matches = bf.match(des1, des2)
matches = sorted(matches, key=lambda x: x.distance)

# Display best matches
result = cv2.drawMatches(
    ref, kp1, test, kp2,
    matches[:30], None,
    flags=cv2.DrawMatchesFlags_NOT_DRAW_SINGLE_POINTS
)

print("Number of matching features:", len(matches))

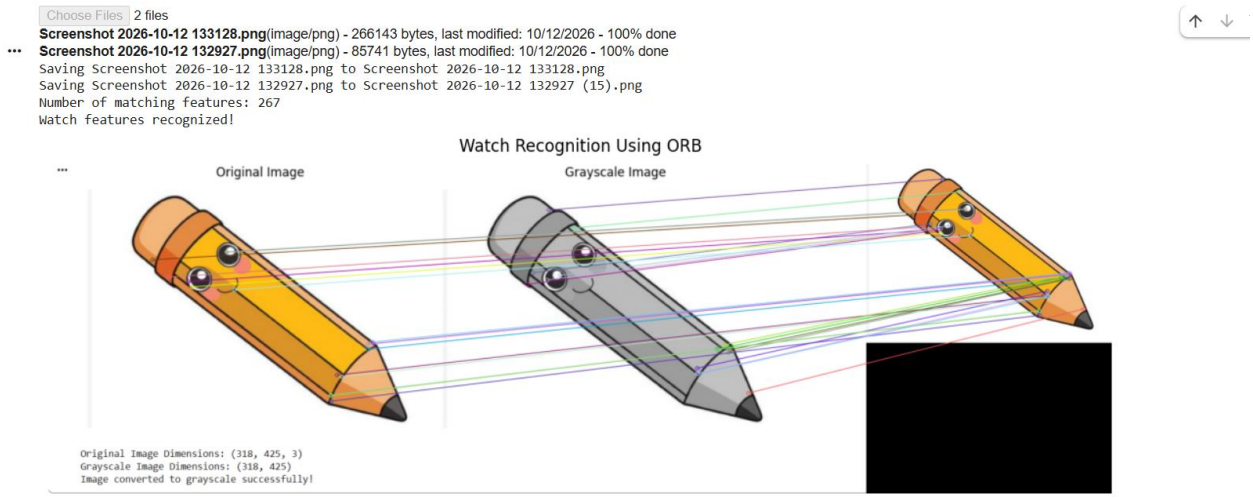
if len(matches) >= 10:
    print("Watch features recognized!")
else:
    print("Watch not confidently recognized.")

plt.figure(figsize=(14, 6))
plt.imshow(cv2.cvtColor(result, cv2.COLOR_BGR2RGB))
plt.title("Watch Recognition Using ORB")
plt.axis("off")
plt.show()
```

Expected Output

The output displays matching feature points between the reference watch and the test image.

4. Output



The watch features are detected and matched with the reference image using ORB feature matching.

5. Result

Thus, watch recognition using ORB feature detection and matching was successfully implemented using Python and OpenCV.